

Galactic Plane Rotation Report

Galactic and Extragalactic Lab



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Abstract

Neutral hydrogen (HI) 21 cm line observations are a useful way to study the structure and motion of gas in the Milky Way. For this project, the Tony Fairall Teaching Telescope was used to observe 21 cm line emission at three Galactic longitudes ($l = 22^\circ$, 276° , and 324°) along the Galactic plane. After calibration and analysis of the spectra, the maximum relative velocities were found to be $v = 89.52 \pm 9 \text{ km s}^{-1}$ (redshifted) at $l = 22^\circ$, $v = 0 \pm 6.3 \text{ km s}^{-1}$ at $l = 276^\circ$, and $v = -51.52 \pm 11 \text{ km s}^{-1}$ (blueshifted) at $l = 324^\circ$. These results fit the general trend expected from Galactic rotation, with redshifts and blueshifts on opposite sides of the Galactic Centre, and may also include contributions from local gas motions.

The velocities were calculated using the non-relativistic Doppler equation on the calibrated frequency spectra, then adjusted to the Local Standard of Rest. The uncertainties shown come from noise in the spectra and Gaussian fitting. Additional errors probably exist due to instrumental issues. In particular, software faults during observations caused some data to be lost, and one target was observed at different times of day, which may have affected the calibration. Even with these issues, the observations clearly show how HI 21 cm spectroscopy can be used to trace Galactic kinematics and why careful calibration and consistent observing are important.

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Introduction

Background

Neutral hydrogen (HI) is a key tracer of the large-scale structure of the Galaxy. It allows the measurement of gas velocities via Doppler shifts and provides insight into magnetic field strengths through Zeeman splitting. The hydrogen atom in its ground state exhibits hyperfine structure arising from the magnetic interaction between the proton and electron spins. When the spins flip from parallel to antiparallel alignment, a photon with a wavelength of 21 cm is emitted. Although the transition is intrinsically rare, the high abundance of hydrogen in the interstellar medium ensures that 21 cm emission is widespread and can be observed throughout the Galactic plane [1].

Mapping the 21 cm line across different Galactic longitudes allows astronomers to construct the Milky Way's rotation curve. This reveals the mass distribution in the Galaxy, including contributions from dark matter in the disk and halo [2]. the Galactic disk and halo[2].

Mathematical Reasoning

The basic principle behind Galactic rotation measurements is to use the Doppler shift of the 21 cm line to determine the line-of-sight velocity of the gas relative to the observer. The velocity is computed from the difference between the observed frequency and the rest frequency using the non-relativistic Doppler formular:

$$v_{\text{rad}} = c \frac{f_{\text{rest}} - f_{\text{obs}}}{f_{\text{rest}}}, \quad (1)$$

[3]

whereby c is the speed of light and f_{rest} is the frequency corresponding to the 21 cm emission. By observing directions along the inner Galaxy ($0^\circ < l < 90^\circ$ and $270^\circ < l < 360^\circ$), the maximum radial velocity along each line of sight corresponds to the *tangent point*, the closest approach of the line of sight to the Galactic centre.

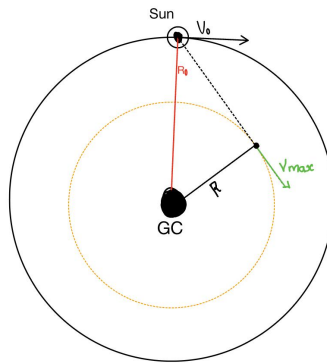


Figure 1: Caption

From this maximum velocity v_{max} , one can infer the minimum galactocentric radius R_{min} using

$$R_{\min} = R_0 \sin l, \quad (2)$$

where $R_0 = 8.5$ kpc is the Sun's distance from the Galactic centre and l is the Galactic longitude. To obtain accurate velocities, it is necessary to correct for the Earth's motion relative to the *Local Standard of Rest* (LSR). The LSR correction accounts for the Sun's peculiar velocity and the Earth's orbital motion, ensuring that derived velocities reflect the motion of interstellar gas with respect to a common Galactic frame.

Observations and Data

Observational Methodology

The Tony Fairall teaching Telescope with site longitude 33.933° and site latitude 18.476° was first pointed towards 3 different Galactic longitudes ($l = 22^\circ$, 276° and 324°) along the Galactic plane ($b = 0^\circ$). The receiver was tuned to a center frequency near 1428.75 MHz with a bandwidth of 6.25 MHz to include both redshifted and blueshifted 21 cm line emission. This ensured that the hydrogen line from the chosen Galactic direction fell within the spectrometer band.

Observations were conducted in an ON-OFF mode by alternating between the target position and a nearby reference position. Subtracting the OFF scan from the ON scan removes the background system noise and instrumental baselines, leaving only the astronomical signal. A dedicated calibration scan of the standard source S8 was then performed. This reference source was used to convert the measured detector counts (ADU) into absolute brightness temperature by comparing the observed S8 spectrum with the corresponding profile from the Leiden/Argentine/Bonn (LAB) all-sky HI [3].

This step ensures that the final spectra can be expressed in physical temperature units. Multiple repeated scans for each pointing were stacked to increase the signal-to-noise ratio. A low-order polynomial baseline was fitted to line-free regions and subtracted to remove instrumental drifts and continuum emission. This produced cleaner spectra that more accurately reflect the HI line shape. The frequency axis for each spectrum was derived using the base frequency and channel spacing contained in the FITS header. In cases where the header centre frequency was offset, the appropriate corrections were applied. Constructing an accurate frequency axis is essential for reliable velocity measurements.

The observed frequency axis was then shifted into the Local Standard of Rest (LSR) frame using the telescope location and the time of observation. This correction removes the effect of the Earth's rotation and orbital motion, as well as the Sun's peculiar motion, allowing the spectra to be placed in a common Galactic velocity reference frame. The S8 calibration was applied by fitting a linear relation between the stacked S8 spectrum in ADU and the LAB survey brightness temperature in Kelvin [4][3]. The resulting calibration factor and offset were applied to all target spectra to convert them into brightness temperature. This step ensures that the observed intensities can be physically interpreted and compared across different observations.

Finally, the corrected frequency axis was converted to radial velocity using the non-relativistic Doppler formula and the 21 cm rest frequency ($f_{\text{rest}} = 1420.40575177$ MHz). Converting to velocity space allows the identification of maximum line-of-sight velocities, which are used to determine the galactocentric radii and trace the rotation curve of the Milky Way.

Data Filtration and Analysis

Time (UTC)	Galactic Lat (b)[°]	Galactic Long (l)[°]	RA[°]	Dec[°]
2025-09-16 13:46:37	0	22	18.52	-9.70
2025-09-16 09:09:29	0	276	9.65	-52.51
2025-09-16 09:56:19	0	324	15.54	-56.16

Table 1: Observation Log of the 3 chosen targets using the TFFT at the Observatory in Capetown South Africa at 2 separate times of the day as shown in the time difference, also showing their coordinates on the sky (celestial sphere).

Using this information and the radio telescope and python code to plot the on off reading, we have the following power spectra for all 3 targets :

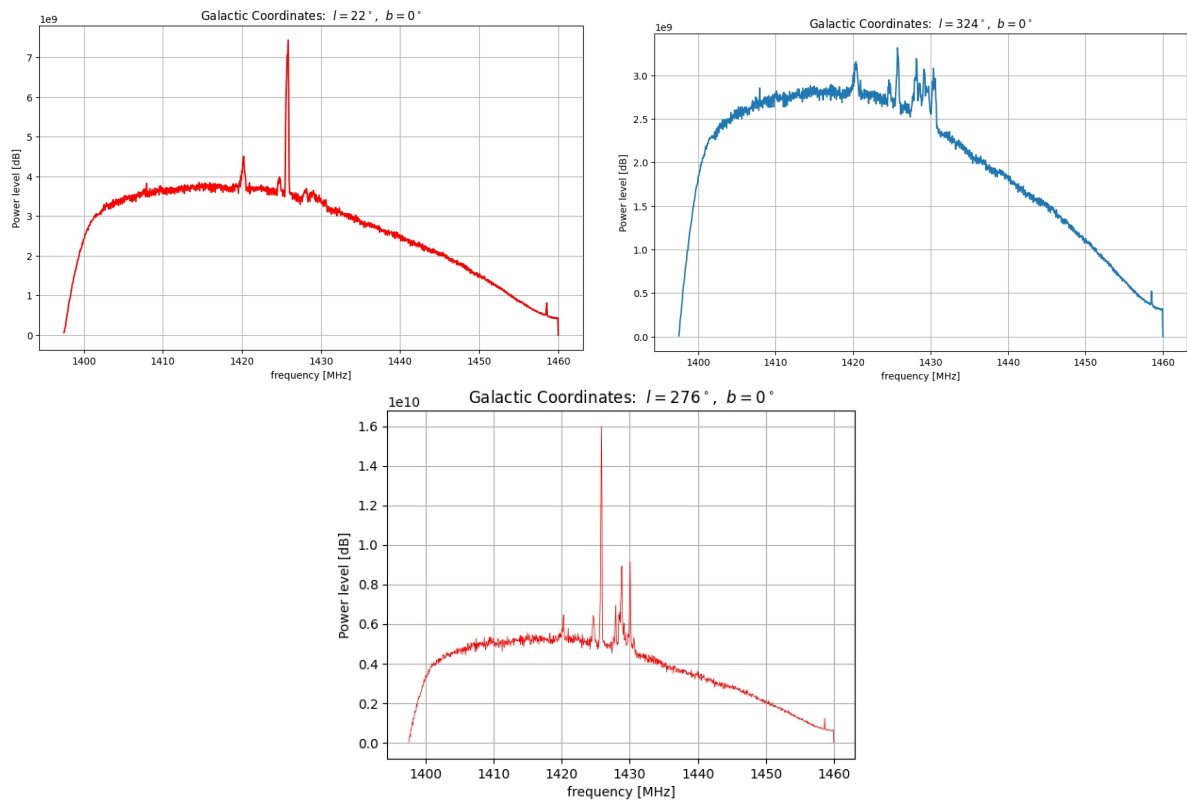


Figure 2: The three figures were taken using the left dish and is the stacked using the accuried fits files, the number of observation varies due to the softwear shutting down mid observing and the Frequency axis was adjusted to the correct range as suggested in the manual

Isolation of the range of interest of the frequency axis around the rest frequency of HI and removal of the noise the data from the raw stacked spectrum gives us a visible picture of the features and we get the following

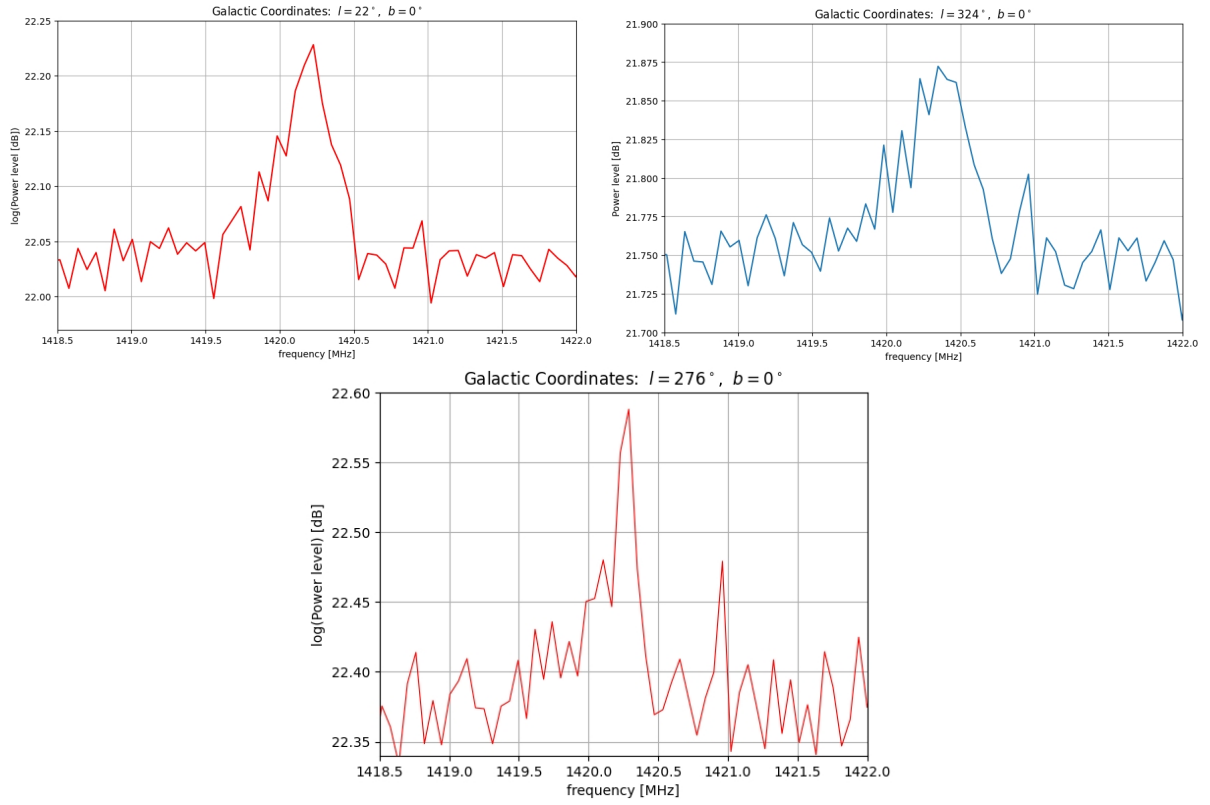


Figure 3: The 3 figures show the features of interest around 1420.40575177 MHz in log scale for the y-axis with the noise cut out.

The final calibrated spectrum for the observations of all the targets represented below :

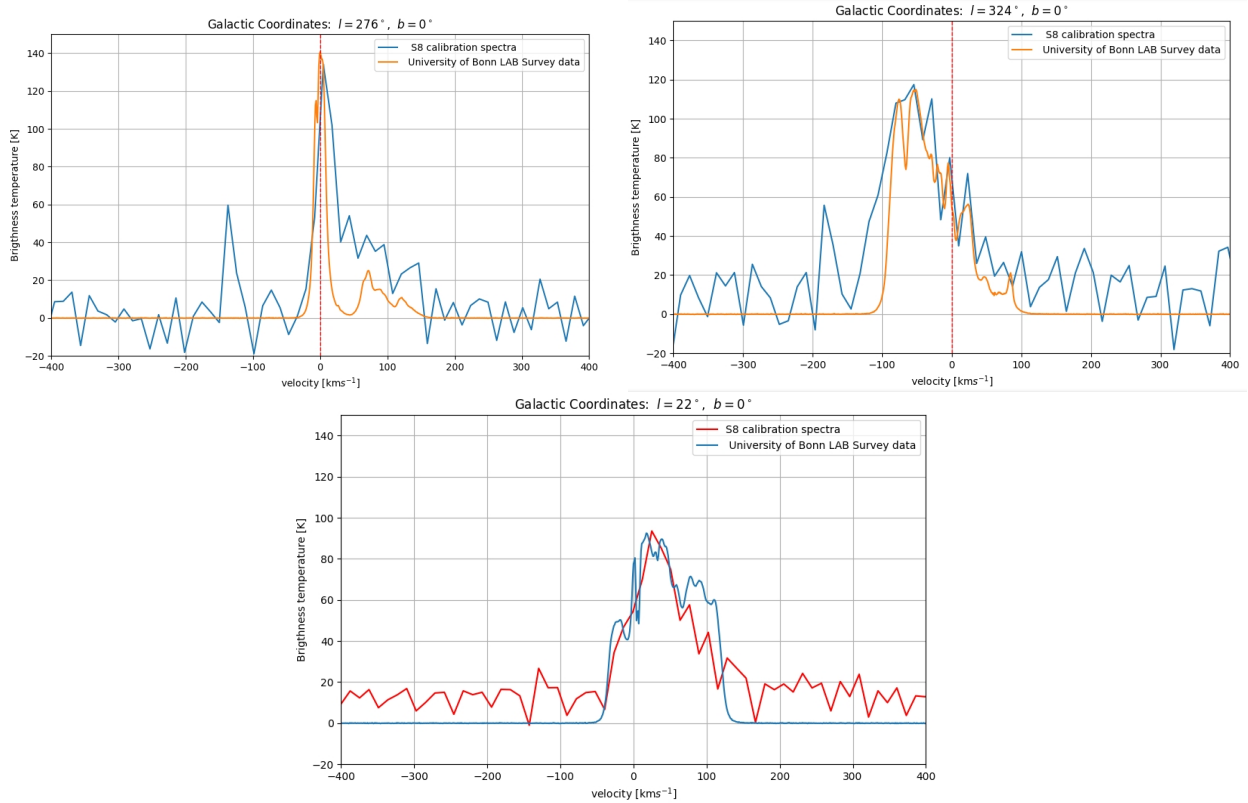


Figure 4: calibrated spectrum for targets with data from the LAB survey labeled mistakenly as the 'S8 calibration spectra'

From these calibrated figures we attained the maximum relative velocities, $v = 0 \pm 6.3 \text{ km.s}^{-1}$ ($l = 276^\circ$) showing that the target is almost not red or blue shifted, $v = -51.52 \pm 11 \text{ km.s}^{-1}$ ($l = 324^\circ$) showing that in this and $v = 89.52 \pm 9 \text{ km.s}^{-1}$ ($l = 22^\circ$) and also using the equation $R_{min} = R_0 |\sin(l)|$ whereby the R_0 used is the value found in the manual without uncertainty equaling 8.5 [3].

l [$^\circ$]	Blue or red shift	R_{min} [kpc]	v_{max} [km.s $^{-1}$]
22	redshifted	3.18	89.5 ± 9.0
276	almost None	8.45	0.0 ± 6.3
324	Blue shifted	5.00	-51.5 ± 11.0

Table 2: summary of results obtained from the observational analysis for three sources.

Discussions and Interpretation

Interpretation of the Measured Velocities

The measured radial velocities at different Galactic longitudes provide insight into the kinematic behaviour of the observed regions. At $l = 276^\circ$, the measured velocity is $v = 0 \pm 6.3 \text{ km s}^{-1}$, which is effectively consistent with zero within the uncertainties. This indicates that the target shows no significant Doppler shift, meaning it is neither approaching nor receding relative to the observer. At $l = 324^\circ$, the measured velocity is $v = -51.52 \pm 11 \text{ km s}^{-1}$.

The negative sign indicates a clear blueshift, signifying that the source is moving toward the observer along the line of sight. The detection is significant at approximately 4.7σ , which makes this a robust indication of approaching motion. Finally, at $l = 22^\circ$, the measured velocity is $v = 89.52 \pm 9 \text{ km s}^{-1}$, indicating a strong redshift. This corresponds to the source moving away from the observer, with a significance level of nearly 10σ , representing a very strong detection of recession.

Plausible Physical Causes

The observed pattern of redshifted and blueshifted velocities can be attributed to several physical causes. Firstly, Galactic differential rotation naturally produces opposite signs of radial velocities on opposite sides of the Galactic [1]. The negative velocity at $l = 324^\circ$ and the positive velocity at $l = 22^\circ$ are roughly symmetric in longitude, consistent with the expected kinematic signature of Galactic rotation.

Secondly, local peculiar motions such as streaming flows along spiral arms, expanding or contracting shells, and large-scale motions within the interstellar medium can contribute to deviations from purely circular rotation, and may account for the relatively large absolute velocities observed. Additionally, distance ambiguities inside the solar circle can complicate the interpretation, as a single velocity can correspond to both near and far kinematic distances. Finally, heliocentric velocities can differ from Local Standard of Rest (LSR) velocities by approximately $10\text{--}20 \text{ km s}^{-1}$ due to the Sun's peculiar motion, and applying this correction is essential for accurate comparison with Galactic rotation models.

Uncertainties and Observational Limitations

The uncertainties reported for the measured velocities correspond to statistical errors derived from the spectral line fitting process. These uncertainties primarily reflect the noise level in the spectra and the fitting residuals after applying a Gaussian model to the observed line profiles. However, the overall uncertainty in the results is likely larger due to several systematic factors. Firstly, the data acquisition process was affected by software faults, which resulted in incomplete or corrupted datasets for certain observations.

This limited the amount of reliable data available for analysis and may have introduced additional, unquantified errors into the velocity measurements. Secondly, for one of the target sources, measurements were taken at different times during the day. Variations in instrumental calibration and environmental conditions (such as temperature changes affecting receiver stability) between these sessions could have contributed to small systematic shifts in the measured line positions. These factors highlight that while the quoted uncertainties provide an estimate of the random

errors, they do not fully capture the potential systematic uncertainties arising from data quality issues and observational conditions.

Bibliography

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